

TODO: manuscript title

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0.1 Introduction

5 TODO: Write introduction.

0.2 Methods

Source: [Article Notebook](#)

0.2.1 Vegetation and palaeoclimate data

We sourced all vegetation community and climate records from **VegVault** (version 1.0.0;
10 [cite]), a harmonised SQLite database that integrates fossil pollen archives and contemporary vegetation datasets with co-registered palaeoclimate reconstructions and plant functional traits. Sourcing all three data types required by this study — fossil and contemporary vegetation community records, co-registered palaeoclimate predictors, and plant functional traits — from a single harmonised repository offers two critical advantages. First, it guarantees
15 consistent site identifiers and temporal grids across all inputs, which is a prerequisite for the aligned multi-species modelling framework used here. Second, because VegVault is versioned and publicly archived, the exact set of records used in this study can be retrieved by any re-

searcher with access to the same database version, making the data inputs fully reproducible.

Fossil pollen records are drawn from the Neotoma Paleoecology Database^{Williams2018?} and

20 FOSSILPOL [cite], and contemporary records are drawn from modern vegetation databases BIEN [cite] and sPlotOpen [cite]. Plant functional traits were assembled in VegVault from TRY^{Kattge2020?} and additional contributed datasets, with the original dataset and data-source identities retained for provenance and correction. Palaeoclimate predictors are CHELSA-TraCE21k bioclimatic variables^{Karger2021?} co-registered in VegVault at the site
25 and time-step level.

We selected seven CHELSA bioclimatic variables as candidate abiotic predictors: BIO1 (mean annual temperature), BIO4 (temperature seasonality), BIO6 (minimum temperature of the coldest month), BIO12 (annual precipitation), BIO15 (precipitation seasonality), BIO18 (precipitation of the warmest quarter), and BIO19 (precipitation of the coldest quarter). These
30 variables represent the primary dimensions of climate space known to govern plant distributions — mean thermal energy, climatic extremes, and water availability across seasons — while omitting correlated secondary variables to limit redundancy before formal collinearity screening. Full descriptions and units of all bioclimatic variables are provided in Supplementary Table S[TODO].

35 The geographic and temporal coverage of extracted records (number of fossil pollen and contemporary sites per 500-year time slice and per continental region) is summarised in Supplementary Figure S[TODO].

0.2.2 Taxonomic harmonisation

Fossil pollen records originate from multiple source databases that use different original
40 nomenclature systems and taxonomic conventions; a unified classification is therefore a pre-
requisite for any cross-database comparison. The guiding principle of the harmonisation was
to resolve each pollen morphotype to the **finest taxonomic rank the morphological evi-
dence supports**, rather than imposing a fixed target rank across all morphotypes. For each
morphotype, we retrieved taxonomy with `taxospace::get_classification()` against the
45 GBIF taxonomic backbone [cite], using non-exact matching where needed to resolve legacy
or region-specific names while preserving traceability of the returned identifiers. Where the
morphotype was too broadly defined morphologically to support a confident genus assignment
— typically for compound types that aggregate morphologically similar pollen from several
genera — the record was retained at family rank rather than discarded, preserving ecological
50 information that would otherwise be lost entirely. We excluded non-plant records (e.g., fungal
spores and fern spores flagged in source databases). We supplemented the backbone matching
with a curated project-specific correction table that records manually validated taxonomic
decisions for morphotypes known to be mis-classified or absent from the backbone, ensuring
that recurring edge cases are resolved consistently across all analyses. All classifications were
55 reviewed and validated by expert botanists to catch systematic errors arising from regional
nomenclatural traditions or legacy morphotype names not covered by the backbone. Mor-
photypes that remained unresolved after automated matching and manual correction were
exported as explicit missing-taxon reports and tracked separately from the final harmonised
tables. Because the taxa present in any given analysis depend on the geographic extent and

temporal window queried, we compiled a separate harmonisation table for each continental region rather than a single global table. We provide these tables as Supplementary Tables S[TODO] (one per continental region).

0.2.3 Multi-scale analytical framework

Co-occurrence patterns are shaped by processes operating at different spatial extents, across geological time, and at different levels of biological organisation. Because the relative importance of these processes is itself expected to vary across scales, pooling all data at a single arbitrary scale risks masking or confounding mechanisms that are only visible at specific extents or resolutions. We therefore applied the same modelling and analysis workflow at multiple levels along three axes: **spatial** (continental, regional, and local extents), **tempo-**
ral (40 successive 500-year slices from the Last Glacial Maximum to the present), and **taxo-**
nomic (genus, family, and functional-type level). We first describe the shared pre-processing and modelling steps common to all analyses, then detail how each axis was defined and the reasoning behind specific choices.

0.2.3.1 Shared pre-processing

Community data preparation. We applied several pre-processing steps to the raw community matrices to prepare them for modelling and ensure comparability across analyses. We converted raw pollen counts to proportional abundance (relative pollen percentage). To standardise temporal resolution across irregular sampling intervals (typical for palaeoecological datasets), we then linearly interpolated records to a common 500-year temporal grid. We considered binning but preferred interpolation to preserve the full temporal resolution

of the data and avoid artefacts introduced by arbitrary bin edges. Temporal interpolation was applied only to fossil sequences; contemporary samples were retained as present-day observations at age 0. We excluded taxa contributing less than 1 % of the pollen sum within any individual sample from that sample’s record, because such occurrences are below the
85 detection limit reliably resolvable from pollen counts and contribute predominantly noise to co-occurrence estimates. We used presence/absence rather than continuous abundance as the response variable, because pollen abundances are strongly influenced by taxon-specific pollen production rates and dispersal distances — biases that are effectively removed by binarisation, leaving a response that more directly reflects co-occurrence [TODO: cite]. Before
90 model assembly, we restricted analyses to the intersection of site-age samples present in community, coordinate, and abiotic tables, and required abiotic records to be complete across all retained abiotic variables. Finally, to ensure sufficient data density for reliable model estimates, we excluded spatial units with too few sites or too few qualifying taxa after filtering. We calibrated the exact thresholds separately for each spatial unit, because the number of
95 sites and taxa varies across units and the optimal thresholds depend on the data structure of each unit. For contemporary vegetation records, we applied an additional preprocessing stage before shared filtering: we generated a quality-control report to flag impossible values (e.g., non-zero modern ages, invalid coordinates, and missing or negative abundance values), removed exact duplicate records deterministically, and aggregated colocated records only
100 within eligible same-prefix source groups at identical coordinates and age. Cross-database colocations (notably BIEN/sPlot overlaps) were retained unchanged and tracked separately for auditability. For aggregated colocations, we replaced original records with synthetic site identifiers and propagated aggregation consistently across community, abiotic, and coordi-

nate tables via explicit mapping tables.

105 **Abiotic predictor selection.** Collinear predictors inflate the variance of coefficient estimates and may result in overfitting, because variance attributable to one predictor is spuriously shared with its collinear counterparts. We therefore screened candidate bioclimatic variables for redundancy using a dual-filtering approach that combines pairwise Pearson correlation and Variance Inflation Factor (VIF) [cite Benito 2025]. We calibrated both thresholds
110 automatically to the correlation structure of each specific dataset rather than setting fixed global values, because optimal thresholds depend on the number and collinearity of predictors, which vary across spatial and temporal units. When two predictors were redundant, we preferentially retained the one with greater univariate predictive power, ensuring that the non-redundant subset maximises ecological information.

115 **Variable scaling.** We centred and z-score scaled bioclimatic predictors to a common variance, which is required for stable gradient-descent optimisation and allows coefficient magnitudes to be compared across predictors with different natural units. Rows with missing values in the abiotic predictor table were removed prior to scaling, so all scaled predictors supplied to the model were complete-case. In analyses with age as a predictor, we centred age but
120 did not variance-scale it, so that interaction coefficients retain their natural interpretation in units of years. We projected site coordinates to a metric coordinate system (ETRS89-LAEA, EPSG:3035) so that geographic distances are measured in kilometres rather than decimal degrees.

Moran Eigenvector Maps (MEMs). Spatial autocorrelation in species distributions

125 arises from unmeasured environmental gradients, dispersal limitation, and historical contin-

gency — processes that are not captured by the bioclimatic predictors but would inflate the biotic residual (B fraction) of the variance decomposition if left unaccounted for. We represented large-scale spatial structure using Moran Eigenvector Maps (MEMs), which are eigenvectors of a spatial weights matrix and span the full range of positive spatial autocorrelation present in the data [cite]. Including MEMs as predictor variables in the spatial component of the model approximates a spatial random field without the computational cost of full Gaussian process random effects, which would be prohibitive for the large community matrices. For purely spatial analyses, we derived MEMs from a two-dimensional geographic distance matrix among sites. For spatiotemporal analyses, we computed MEMs on a sample-level three-dimensional coordinate matrix (x, y, age), where age was converted from years to kiloyears and each axis was z-scored before eigenvector extraction so that spatial and temporal extents contributed on comparable scales. In both spatial and spatiotemporal settings, if the requested number of MEM predictors exceeded the number of positive eigenvectors returned by the decomposition, we clamped retention to the available positive set.

0.2.3.2 Shared modelling workflow

Joint species distribution model. We modelled plant co-occurrence using a Joint Species Distribution Model (jSDM)^{Warton2015?}, which fits all species simultaneously within a single likelihood framework and estimates the residual covariance among species after accounting for shared environmental and spatial responses. We chose the jSDM framework over single-species distribution models because it explicitly models the residual inter-specific covariance — that is, co-occurrences that cannot be attributed to shared responses to environment or space — which is the central quantity of interest in this study. We performed all analyses

using the R package {sjSDM}^{Pichler2021?}, which fits jSDMs via stochastic gradient descent with an optional GPU backend, making it feasible for the large community matrices (up to
150 several hundred taxa $\times$ thousands of sites) analysed here.

Each taxon’s occurrence was modelled as the linear combination of three additive components:

- **Environment (A)** — the non-collinear bioclimatic predictors, with age $\times$ climate interaction terms to allow the strength of climate filtering to vary across the Last
155 Glacial–Interglacial cycle. Only the interaction of age with climate predictors was included; the direct age term was excluded to avoid absorbing broad temporal trend into the environment component.
- **Space (S)** — MEM predictors included as fixed effects to absorb spatially (or spatiotemporally) structured variation not explained by the climate predictors, thereby
160 preventing unmeasured spatial processes from inflating the biotic residual.
- **Biotic residual (B)** — a shared low-rank latent covariance matrix that captures the residual pairwise associations among taxa after accounting for their shared responses to environment and space. Positive residual covariance indicates that two taxa co-occur more often than expected given environment and spatial structure; negative covariance
165 indicates mutual exclusion.

Model fit and convergence. Because the primary goal of this study is variance decomposition rather than out-of-sample prediction, we assessed model performance on the training data using Nagelkerke R^2 and per-species AUC, accuracy, and log-loss. Convergence was confirmed by inspecting the training loss trajectory: we considered a model converged when

the linear trend of the loss curve was negligible and successive steps produced only marginal changes, indicating that further optimisation would not meaningfully alter the parameter estimates.

Variance decomposition. To attribute the variance in community composition to ecologically interpretable sources, we applied a Monte Carlo ANOVA decomposition to each fitted jSDM. The approach repeatedly draws from the fitted model with one or more components omitted and quantifies how much variation is lost, yielding seven non-overlapping Venn-diagram fractions:

- **A** — variance uniquely attributable to the abiotic (environment) component;
- **S** — variance attributable to the spatial component (MEMs);
- **B** — variance attributable to the residual biotic covariance (co-occurrence patterns unexplained by environment or space, interpreted as a signal of potential biotic interactions^{Warton2015?});
- **AB, AS, BS, ABS** — variance jointly attributable to pairs or all three components.

We set fractions that were negative — an artefact of Monte Carlo approximation error — to zero, as negative explained variance has no ecological interpretation. For all temporal summaries and cross-model comparisons, we then re-expressed the seven-fraction output as three unique components (Abiotic, Spatial, Associations) using a Shapley equal-split allocation of shared fractions (AB, AS, BS, ABS) among their contributing components. Concretely, AB and AS are split evenly between their two parent components and ABS is split one-third to each of A, S, and B, yielding adjusted unique contributions that sum to total explained variance. We finally converted these adjusted components to percentages of explained vari-

ance within each fitted model, and these percentage trajectories are the quantities reported in temporal comparisons.

0.2.3.3 Spatial axis

To isolate the effect of geographic extent on co-occurrence patterns, we applied the shared workflow independently at three nested spatial scales while keeping the temporal and taxonomic dimensions constant. The three scales were chosen to span the range from macroecological processes — where broad climatic gradients dominate co-occurrence — to local processes, where dispersal limitation and microhabitat filtering are more prominent:

- **Continental** — three units: Europe (-10° to 40° E, 35° to 70° N), North America (-130° to -60° W, 30° to 70° N), and northern Asia (60° to 140° E, 50° to 75° N).
- **Regional** — approximately 30 geographic units per continent, each spanning roughly 20° longitude $\times$ 10° – 15° latitude.
- **Local** — approximately 100 fine-grain units per continent, each spanning roughly 5° longitude $\times$ 5° latitude.

We compared variance fractions across scales to identify which components of co-occurrence signal are scale-invariant and which are scale-dependent.

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Three scale-specific thresholds governed which spatial units were retained for model fitting.

We excluded a taxon from a spatial unit if it appeared in fewer than 5 distinct sediment cores.

We further excluded taxa if they appeared in fewer than 10 distinct site–age combinations at

the continental scale, fewer than 10 at the regional scale, or fewer than 5 at the local scale, reflecting the reduced number of sites per unit at finer grains. We excluded a spatial unit entirely if fewer than 5 qualifying taxa remained after all pre-processing at the continental
215 scale; the corresponding lower bounds were 4 taxa at the regional scale and 3 taxa at the local scale.

0.2.3.4 Temporal axis

To test whether the structure of plant co-occurrence has changed through time — and in particular whether the large climate changes since the Last Glacial Maximum are associated with
220 a reorganisation of co-occurrence networks — we applied the shared workflow independently to each 500-year time slice across the full 20,000-year window. We fitted a separate jSDM to each time slice rather than a single pooled model with a temporal random effect. Prior to model fitting, we computed co-occurrence network metrics (connectance, nestedness, and modularity) directly from the binarised community matrix at each time slice to characterise
225 the structural properties of the co-occurrence network independently of any model assumptions. We then aggregated results from successive time slices into temporal trajectories of variance fractions and network metrics, revealing how the relative importance of environment, space, and biotic residuals — and the structure of the co-occurrence network itself — has evolved over geological time. We ran analyses for three continental regions: Europe, North
230 America, and northern Asia.

Source: [Article Notebook](#)

We applied the same three data-quality thresholds at each time slice within each continental

region. We excluded a taxon from a given time slice if it appeared in fewer than 5 distinct sediment cores, or in fewer than 10 distinct site–age combinations. We excluded a time slice
235 entirely if fewer than 5 qualifying taxa remained after all pre-processing.

0.2.3.5 Taxonomic axis

The level at which taxa are defined determines which ecological processes are legible in co-occurrence patterns: genus-level co-occurrence may reflect fine-grained competitive or facilitative interactions, whereas broader groupings may reveal assembly rules driven by shared
240 functional strategies. To test how strongly co-occurrence patterns depend on the level of biological organisation, we collapsed the community matrices to three taxonomic resolutions: **Genus**, **Family** and **Functional type (FT)** — ecologically coherent groups derived from functional trait clustering (see *Functional-type classification* below).

We applied the same jSDM pipeline at each resolution without modification, enabling direct
245 comparison of variance fractions across taxonomic scales. We evaluated the taxonomic axis only at the spatial scale, fitting each resolution independently per spatial unit.

0.2.4 Functional-type classification

Trait data, correction and audit. Plant functional traits characterise the ecological strategies of taxa and provide a biologically grounded basis for grouping species into func-
250 tional types that reflect shared responses to the environment rather than arbitrary taxonomic boundaries. Trait processing followed seven ordered steps.

1. **Extraction and provenance.** We extracted six continuous trait domains from Veg-

Vault 1.0.0: diaspore mass, leaf area, leaf mass per area, leaf nitrogen content per unit mass, plant height, and stem specific density (see Supplementary Table S[TODO] for definitions and units). The extraction retained dataset, data-source, sample, trait, and taxon identifiers so that every correction could be restricted to its documented source and audited at record level. We removed records without a taxon identifier or with a non-finite trait value.

2. **Source-level unit harmonisation.** Starting from the otherwise unchanged extraction, we applied a version-guarded compatibility layer before any taxon-level review. The 10 approved rules corrected 28,856 records by multiplying affected leaf-area and plant-height sources by 100 and the affected stem-density source by 0.001. Each rule specified the VegVault version, source identifier and description, trait domain, factor, and expected match count. The pipeline stopped if the database version, source identity, selector, or match count differed from the approved contract. Record-level audit tables retained the original value, corrected value, factor, and source identifiers.
3. **Raw review-candidate detection.** We constructed one candidate for each taxon-trait-domain group containing a non-positive value, a within-taxon outlier, or a key preserved from the archived trait review. A within-taxon outlier was a value whose absolute deviation from the group median exceeded 1.5 times the interquartile range; this test was applied only to groups with at least 10 records and a non-zero interquartile range. We also calculated a whole-domain diagnostic fence of 3 interquartile ranges to identify possible source-wide problems. Whole-domain flags informed source investigation but did not create taxon-level review candidates by themselves.

275 4. **Raw evidence review and correction.** Each candidate received an atomic decision
of no action, exclusion, or multiplication by a positive factor. Correction decisions could
be restricted by exact trait name, dataset, and inclusive or exclusive value bounds. Evi-
dence came from the archived review, deterministic distributional summaries, grouped-
source comparisons, and bounded AI-assisted evidence gathering. AI-generated rec-
ommendations remained proposals; only decisions explicitly approved by the project
280 owner with reviewer and date metadata entered the pipeline. The completed raw gate
contained 4,339 approved decisions: 4,262 no-action decisions, 20 exclusions, and 57
residual scaling rules. Before applying these decisions, validation rejected incomplete
coverage, unmatched selectors, overlapping or conflicting rules, and any taxon-level
285 scaling that would duplicate or contradict source-level scaling.

5. **Taxonomic resolution and second review.** After applying raw decisions, we clas-
sified trait taxa against the same taxonomic backbone and curated auxiliary table used
for community records, then resolved each trait name to its finest supported rank.
We rebuilt the candidate set under the resolved names because merging synonymous
or differently resolved names can reveal distributions that were not apparent before
290 harmonisation. The classified gate contained 4,223 approved no-action decisions, con-
firming that no additional correction was required after source scaling, raw review, and
taxonomic resolution.

6. **Matching and aggregation.** For each continental functional-type analysis, we
295 matched corrected trait records to the canonical taxonomic lineage of each community
taxon. A community taxon resolved above species level therefore received observa-

tions from all matching descendant trait taxa. We calculated the median for each community-taxon-trait-domain combination and pivoted the result to a taxon $\times$ trait matrix.

300 **7. Interim release and uncertainty handling.** We froze the resulting records as an interim analysis release specific to VegVault 1.0.0. A future VegVault version intentionally fails this compatibility contract until the local source rules are explicitly retired or replaced. All six traits were retained in the primary analysis, but leaf mass per area and leaf nitrogen content per unit mass were marked as having provisional unit provenance, and stem specific density was marked as retaining interim source uncertainty.
305 Leaf mass per area received no scaling or exclusion because the available evidence did not justify a correction, rather than because its source units were considered resolved. We therefore defined four separately stored sensitivity classifications that omitted leaf mass per area, leaf nitrogen content, both provisional traits together, or stem specific
310 density.

Distance computation and clustering. We derived functional types separately for each continental unit (Europe, North America, northern Asia) to ensure that the groupings reflect the functional diversity actually present in each region rather than a globally imposed classification. Before calculating distances, we log10-transformed diaspore mass, leaf area,
315 leaf mass per area, and plant height to reduce the influence of their strongly right-skewed measurement scales; leaf nitrogen content and stem specific density were retained on their original scales. We quantified pairwise dissimilarities among taxa using the Gower distance metric, which range-standardises each trait and calculates each taxon pair from their jointly

observed traits without requiring imputation. Pairs sharing no observed trait received the
maximum dissimilarity of one rather than a non-finite distance. We then grouped taxa by
Ward’s minimum-variance hierarchical clustering applied to the Gower distance matrix. We
chose Ward’s method because it minimises within-cluster total variance at each merge step,
producing compact, well-separated clusters that are more likely to correspond to ecologically
coherent functional strategies than linkage methods that optimise only local cohesion.

Selection of the number of functional types. We determined the number of functional
types k by a dual criterion designed to balance cluster quality with data availability for
downstream modelling. For each candidate k in the range $k_{\min}$ ($= 5$) to $k_{\max}$ ($= 30$), we passed
the resulting FT-level community matrix through the same data-filtering steps (rare-type
removal, minimum-site and minimum-sample filters, and binarisation). We considered viable
only k values that left at least a minimum number of non-constant FT columns after filtering,
because a k that produces many rare or invariant functional types provides insufficient data
for stable model estimation regardless of its clustering quality. Among the viable candidates,
we selected the k with the highest mean silhouette width — a model-free measure of cluster
cohesion and separation that does not depend on the clustering algorithm’s internal objective
— as the final number of functional types. This two-stage criterion prevents the selection
algorithm from choosing a fine-grained k that scores well on clustering metrics but collapses
to an inadequate number of informative functional types after real-data filtering. We then
assigned each taxon its functional-type label and collapsed the community matrix to FT level
by taking the union of presences across all member taxa within each site and time point.

0.2.5 Computational reproducibility

Reproducibility of the full analysis was a primary design goal, because the study spans hundreds of independent models across three continents, three spatial scales, and 40 time slices — a scope at which silent failures and undocumented parameter choices are difficult to detect after the fact. We implemented the entire analytical workflow — from data extraction through modelling, variance decomposition, and figure generation — as a declarative, dependency-tracked pipeline using the `{targets}` R package^{Landau2021?}, which caches the output of every computation and automatically re-runs only those steps that are invalidated by an upstream change in code or data. This design makes every result traceable to the exact code and inputs that produced it and ensures that partial re-runs after code edits produce the same output as a full run from scratch. We locked package versions using `{renv}`^{Ushey2023?}, so that the full computational environment — including all R package versions and their dependencies — can be recreated identically on any machine.

Source: [Article Notebook](#)

The analytical codebase comprises 364 modular R functions, each in its own file, spanning data extraction, taxonomic harmonisation, community pre-processing, collinearity screening, spatial eigenvector computation, model fitting, variance decomposition, functional-type classification, and visualisation. Each function has a dedicated unit-test file (399 test files in total), and the test suite achieves 96 % line coverage across all function files as measured by `{covr}` [TODO: cite]. All 364 functions are documented and are publicly accessible through a dedicated project documentation website [GitHub Pages URL; TODO], which provides searchable reference pages for every function alongside narrative descriptions of each analy-

sis phase.

All analysis code, pipeline definitions, configuration files, and instructions for reproducing the environment are publicly available at [GitHub URL / Zenodo DOI; TODO].

0.3 Results

Source: [Article Notebook](#)

0.3.1 Data overview

TODO: Report number of sites, taxa, time windows, and samples retained after filtering (use inline `r` expressions — no hardcoded numbers).

0.3.2 Model performance

TODO: Report predictive performance metrics (AUC, RMSE, etc.).

0.3.3 Co-occurrence patterns

TODO: Describe spatial and temporal patterns in species associations.

0.3.4 Environmental drivers

TODO: Describe ANOVA variance partitioning results.

0.4 Discussion

TODO: Write discussion.

0.5 Acknowledgements

TODO: Add acknowledgements. Funding sources, data providers, computing resources,
380 field/lab assistance.

0.6 Author Contributions

Role	Contributors
Conceptualisation	
Data curation	
Formal analysis	
Funding acquisition	
Investigation	
Methodology	
Project administration	
Resources	
Software	
Supervision	
Validation	
Visualisation	
Writing – original draft	
Writing – review & editing	

0.7 Data Availability

TODO: Describe data availability. Reference VegVault, pipeline outputs, and any archived datasets.

0.8 Code Availability

The analysis code is available at <https://github.com/OndrejMottl/BIODYNAMICS-vegetation-cooccurrence> and archived at TODO (Zenodo DOI to be added upon acceptance).

0.9 References

Supplementary Information

0.10 Supplementary Figures

Figure S1 — TODO title

Source: [Article Notebook](#)

0.11 Supplementary Tables

Table S1 — TODO title

0.12 Supplementary Methods

TODO: Add any extended methodological details not included in the main Methods section (e.g., sensitivity analyses, model diagnostics, additional filtering criteria).